SECTION 12 TRAFFIC SIGNS, ROAD MARKINGS AND ROAD STUDS

PART 1: TRAFFIC SIGNS

GENERAL

Traffic signs 12.04 (5) Traffic signs used in the *works* shall be those manufactured by the Correctional Services Department (CSD) and supplied by the Employer. The *Contractor* shall inform the *Supervisor* of the quantities required 4 months in advance so the *Supervisor* may place the order to the CSD. On intimation of their availability, the *Contractor* shall make arrangements to collect them from Tai Lam Correctional Institution or other CSD facilities as directed by the *Supervisor* and store them at the Site or at the works area. If traffic signs are not available from the CSD, or if the *Supervisor* so directs, they shall be supplied by the *Contractor*. All materials so supplied shall conform to the requirements of this Specification and provision shall be made for them to be tested at the Government laboratory when directed by the *Supervisor*. The *Contractor* shall not be entitled to a compensation event for the completion of the *works* or any *section* thereof, or additional payments, due to the non-availability of traffic signs from the CSD.

(6) Traffic signs shall generally be those manufactured by the Correctional Services Department (CSD). If traffic signs are supplied by the CSD for Works in this Contract, the *Contractor* shall be charged the prices obtainable form CSD. Such charge will be recovered by appropriate deductions from money due to or becoming due to the *Contractor* and shall be excluded from the "Effective Value" for the calculation of Contract Price Fluctuations.

(7) The sign faces shall be so transported as to ensure that they suffer no damage and shall be kept in a dry store until incorporation in the *works*.

(8) The *Contractor* shall notify the *Supervisor* the date he requires to collect the sign faces. Such notification shall be submitted to the *Supervisor* in writing not less than four months prior to the date of the sign face collection.

(9) Where the Contract requires existing sign plates to be replaced by new ones, the *Contractor* shall return the salvaged plates to a Government Store as directed by the *Supervisor*.

MATERIALS

Materials for faces of traffic signs 12.09 (1)***S*** Retroreflective sheeting shall be:

(a) Class Ref. 2 material complying with BS EN 12899: Part 1: 2001, Table 9 for all signs except for those in(b); and

(b) Class Ref. 1 material complying with BS EN 12899: Part 1: 2001, Table 8 for signs for pedestrians and cyclists.

FABRICATION OF TRAFFIC SIGNS

Faces for traffic signs 12.13 (1)***S*** Faces for traffic signs shall be formed using retroreflective or non-retroreflective plastic sheeting. Unless otherwise permitted by the *Supervisor*, a single piece of sheeting shall be used. If more than one sheet is used, the number of sheets shall be kept to a minimum. Only vertical and horizontal joints shall be permitted and all joints in plastic sheeting shall overlap by not less then 6mm. The overlap in the horizontal joints shall be from the top. Butt joint in plastic sheeting shall not be used except between individual planks. Sheeting shall be fixed in accordance with the manufacturer's recommendations.

STORAGE OF MATERIALS

Storage of traffic signs 12.17 (3) Further to Clause 12.17(2), the *Contractor* shall store traffic signs of Class 1 material side by side vertically instead of horizontally and piled up.

PART 2 : ROAD MARKINGS

MATERIALS

Hot-applied thermoplastic material 12.23 (5) Binder for thermoplastic material shall be synthetic hydrocarbon resin or natural resin based. In particular, the binder for alkyd resin thermoplastic road marking material shall consist of maleic-modified glycerol ester of wood resin. The binder shall not contain petroleum based hydrocarbon resins, tall oil resins, blends of tall oil and wood resin or similar derivatives.

Cold-applied preformed material 12.24 (3) (a) Type 'A' – cold setting road marking material

(i) Cold setting road marking material shall be durable and resistant to vehicle fuels and lubricants. The material shall be the "Signalit" manufactured by Dimet (ACI Co.), the road marking compound manufactured by Degussa, or products having equivalent functions or performance as approved by the *Supervisor*.

(ii) Aggregate/filler used shall be light-coloured silica sand, quartz powder, quartz sand, calcined bauxite, calcite or spars.

(iii) The binder shall be methacrylic or acrylic resin

based. The hardeners shall be dibenzoyl peroxide, or other materials as approved by the *Supervisor*.

(b) Type 'B' – retroreflective preformed pliable road marking tape

The cold-applied preformed pliable road marking tape shall be 3M 'Stamark' brand pavement tape or sheeting of series 5730, 350, or 351, or products having equivalent functions or performance as approved by the *Supervisor*.

(4) Cold-applied preformed material shall only be permitted for use as temporary road markings, subject to approval by the *Supervisor*.

LAYING ROAD MARKINGS

Laying of road markings at interface with other contracts

12.33A The *Contractor* shall check the road marking arrangement at the interfaces with other contracts or existing carriageway to ensure that the road markings to be laid will match the designed traffic management measures.

PART 3 : ROAD STUDS

MATERIALS

Road studs

12.42 (5) The *Contractor* shall provide and install reflective road studs in accordance with PS Appendix P.

INSTALLATION OF ROAD STUDS

Shot blasting machine

12.45 Where existing road studs are to be removed by steel shot blasting as directed by the *Supervisor*, the shot blasting machine used shall be the "Blastrac" model 1-14D, or products having equivalent functions and performance as approved by the *Supervisor*. The machine shall be of the self-propelled type, and shall have a minimum cleaning path of 350 mm.

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